

# ESM-SnowMIP reference site simulations

## 1. Introduction

ESM-SnowMIP is aimed at evaluating and improving the modelling of snow as part of the WCRP Grand Challenge “Melting Ice & Global Consequences”. Building on previous SnowMIP initiatives, ESM-SnowMIP complements the LS3MIP subproject of CMIP6 designed to evaluate the land surface modules of Earth System Models and to quantify land-related feedbacks. There are three types of simulations in ESM-SnowMIP, as described in the protocol at <http://www.climate-cryosphere.org/media-gallery/1512-esmsnowmip-simulationprotocol>:

- global coupled simulations with land-surface and ocean models coupled to atmospheric models, as in CMIP6
- global offline simulations with prescribed meteorological forcing for land points only, as in GSWP3
- plot-scale simulations with prescribed meteorological forcing for well-instrumented reference sites, as previously in SnowMIP

The reference site simulations described in this document can also be performed by models that are not set up for the global simulations in GSWP3 or that lack the atmosphere and ocean couplings necessary for CMIP6 simulations.

## 2. Forcing data

Model forcing data can be downloaded from <http://www.geos.ed.ac.uk/~ressery/ESM-SnowMIP.html>. All of the meteorological variables listed in Table 1 are supplied in single files with netCDF and text formats for each site. Tier 1 simulations that all participants should perform use data from in situ measurements, but additional forcing files derived from large-scale and downscaled GSWP3 data will be issued later for optional Tier 2 experiments. The forcing data for all sites have 1-hour timesteps, starting from snow-free conditions in autumn and covering several complete years to allow model spin up.

**Table 1:** Short names in ALMA and CMOR conventions, descriptions and units for meteorological forcing variables.

| Short name |      | Description                          | Units                              |
|------------|------|--------------------------------------|------------------------------------|
| ALMA       | CMOR |                                      |                                    |
| LWdown     | rlds | surface downward longwave radiation  | W m <sup>-2</sup>                  |
| Psurf      | ps   | surface pressure                     | Pa                                 |
| Qair       | hus  | near-surface specific humidity       | kg kg <sup>-1</sup>                |
| Rainf      | pr   | rainfall rate                        | kg m <sup>-2</sup> s <sup>-1</sup> |
| Snowf      | prsn | snowfall rate                        | kg m <sup>-2</sup> s <sup>-1</sup> |
| SWdown     | rsds | surface downward shortwave radiation | W m <sup>-2</sup>                  |
| Tair       | ta   | near-surface air temperature         | K                                  |
| Wind       | ws   | near-surface wind speed              | m s <sup>-1</sup>                  |

The convention for forcing file naming is

`met_forcing_site_start_end.format`

where

`forcing` = `insitu`, `largsc` or `downsc` for in situ, large-scale or downscaled data

`site` = site short name, defined in Section 5

`start` = start year

`end` = end year

`format` = `nc` or `txt` for netCDF or plain text files

e.g. the netCDF file containing in situ forcing data for 1 October 1994 to 30 September 2014 at Col de Porte is called `met_insitu_cdp_1994_2014.nc`. The time variable in the netCDF forcing files is in units of hours since 1900-01-01 00:00:00 and variables are labelled with both the ALMA and CMOR names given in Table 1. Plain text forcing files additionally contain relative humidity RH. Each line of a plain text forcing file has the format

year month day hour SWdown LWdown Rainf Snowf Tair Qair RH Wind Psurf

e.g. the first line of the in situ forcing data plain text file for Col de Porte is

1994 10 1 1 0.0 337.8 0.0e+00 0.0e+00 284.1 8.458e-03 89.2 0.9 86382

### 3. Model output data

Model outputs are preferred in netCDF format with variable names based on those used in CMIP6 and GSWP3. The netCDF files will have a time dimension for all variables and a level dimension for soil state variables. Flux variables listed in Tables 2 and 3 should be averaged over each forcing data timestep and model grid cell, and state variables in Table 4 should be given at the end of each timestep. Model options for subgrid variability, such as tiling, should be disabled for reference site simulations.

**Table 2:** Variable names, descriptions and positive directions for surface energy balance components in model outputs. All are in units of  $\text{W m}^{-2}$ .

| Name  | Description                           | Direction       |
|-------|---------------------------------------|-----------------|
| hfds  | downward heat flux at ground surface  | downward        |
| hfdsn | downward heat flux into snowpack      | downward        |
| hfls  | surface upward latent heat flux       | upward          |
| hfmlt | energy of fusion                      | solid to liquid |
| hfrs  | heat transferred to snowpack by rain  | downward        |
| hfsbl | energy of sublimation                 | solid to vapour |
| hfss  | surface upward sensible heat flux     | upward          |
| rlus  | surface upwelling longwave radiation  | upward          |
| rsus  | surface upwelling shortwave radiation | upward          |

**Table 3:** Variable names and descriptions for water balance components in model outputs. All are in units of  $\text{kg m}^{-2} \text{s}^{-1}$ .

| Name       | Description                                                |
|------------|------------------------------------------------------------|
| esn        | liquid water evaporation from snowpack                     |
| evspsbl    | total water vapour flux from the surface to the atmosphere |
| evspsblsoi | evaporation and sublimation from soil                      |
| evspsblveg | evaporation and sublimation from canopy                    |
| mrrob      | subsurface runoff                                          |
| mrros      | surface runoff                                             |
| sbl        | sublimation of snow                                        |
| snm        | surface snow melt                                          |
| snmsl      | water flowing out of snowpack                              |
| tran       | transpiration                                              |

**Table 4:** Names, descriptions and units for state variables in model outputs.

| Name    | Description                                           | Units              |
|---------|-------------------------------------------------------|--------------------|
| albs    | surface albedo                                        | –                  |
| albsn   | snow albedo                                           | –                  |
| cw      | total canopy water storage                            | $\text{kg m}^{-2}$ |
| lqsn    | mass fraction of liquid water in snowpack             | –                  |
| lwsnl   | liquid water content of snowpack                      | $\text{kg m}^{-2}$ |
| mrfsifr | mass fractions of frozen water in soil layers         | –                  |
| mrlqso  | mass fractions of unfrozen water in soil layers       | –                  |
| mrlsl   | masses of frozen and unfrozen moisture in soil layers | $\text{kg m}^{-2}$ |
| snc     | snow area fraction                                    | –                  |
| snd     | snowdepth                                             | m                  |
| snw     | mass of snowpack                                      | $\text{kg m}^{-2}$ |
| snwc    | mass of snow intercepted by vegetation                | $\text{kg m}^{-2}$ |
| tcs     | vegetation canopy temperature                         | K                  |
| tgs     | temperature of bare soil                              | K                  |
| ts      | surface temperature                                   | K                  |
| tsl     | temperatures of soil layers                           | K                  |
| tsn     | snow internal temperature                             | K                  |
| tsns    | snow surface temperature                              | K                  |

The forcing data files provide precipitation as separate rainfall and snowfall rates. Repartitioning of total precipitation into rain and snow by models is strongly discouraged because this will make mass balance differences between models much harder to interpret. Water balance outputs from models that cannot accept rainfall and snowfall as separate inputs must include the repartitioned rainfall and snowfall rates `Rainf` and `Snowf` in units of  $\text{kg m}^{-2} \text{s}^{-1}$ .

Groups who are not familiar with netCDF may submit results in plain text files. Three files are required for each simulation, with fixed-width columns in the order

`year, month, day, hour, hfds, hfdsn, hfls, hfmlt, hfrs, hfsbl, hfss, rlus, rsus`

for energy balance components,

`year, month, day, hour, esn, evspsbl, evspsblsoi, evspsblveg, mrrob, mrros, sbl, snm, snmsl, tran`

for water balance components, and

`year, month, day, hour, albs, albsn, cw, lqsn, lwsnl, mrfsofr(:), mrlqso(:), mrlsl(:), snc, snd, snw, snwc, tcs, tgs, ts, tsl(:), tsn, tsns`

for state variables, including a separate column for each of the variables defined on soil layers and denoted by `(:)`. Use `-999` for missing variables.

## 4. Simulations

Tier 1 simulations with standard model configurations are to be performed for each site with in situ forcing data. Tier 2 simulations with large-scale and downscaled forcing and model modifications to fix the snow albedo and remove the thermal insulation of snow will be described later. The simulation period should be run at least twice for each site to spin up model state variables, starting from saturated soil conditions and no snow. Outputs should only be returned for the final run.

The requested format for netCDF model output file names is

`model_expt_forcing_site_start_end.nc`

where

`model` = model name

`expt` = REF, FA or NI for reference, fixed albedo or no insulation simulations

e.g. the output file for a CLM reference simulation with in situ forcing from Col de Porte should be called `CLM_REF_insitu_cdp_1994_2014.nc`.

Text output files should have names

`model_table_expt_forcing_site_start_end.txt`

where

`table` = eb, wb or sv for energy balance, water balance or state variable tables

Please direct any questions about the reference site simulations to [snowmip@ed.ac.uk](mailto:snowmip@ed.ac.uk).

## 5. Sites

Table 5 and a csv file (<http://www.geos.ed.ac.uk/~ressery/ESM-SnowMIP/sites.csv>) give characteristics for each of the reference sites. Use nominal values of leaf area index area and height for the grassy sites; snow simulations should not be sensitive to the chosen values. An average snow-free albedo is provided where measurements are available; use a value of 0.2 if none is given. Forcing data for several additional reference sites will be provided when they are available.

### 5.1 BERMS Old Aspen, Saskatchewan, Canada

Even-aged stand of aspen with a thick hazelnut understorey, naturally regenerated after fire in 1919. 10 cm layer of organic litter and peat over a sandy clay loam.

**Table 5.1:** Old Aspen site characteristics

|                                         |                                       |
|-----------------------------------------|---------------------------------------|
| Short name                              | oas                                   |
| Location                                | 53.63°N, 106.20°W                     |
| Elevation                               | 600 m                                 |
| Canopy height                           | 21 m                                  |
| Leaf area index                         | winter stem area ~1, summer 3.7 – 5.2 |
| Snow-free albedo (above canopy)         | 0.14                                  |
| Simulation period                       | 1 October 1997 to 30 September 2010   |
| Temperature/humidity measurement height | 37 m                                  |
| Wind measurement height                 | 38 m                                  |
| Reference                               | Bartlett et al. (2007)                |

### 5.2 BERMS Old Black Spruce, Saskatchewan, Canada

Predominantly black spruce, with some tamarack, jack pine and balsam poplar, naturally regenerated after fire in 1894. Sparse understorey. Peat soil over sandy loam and sand with a raised water table.

**Table 5.2:** Old Black Spruce site characteristics

|                                         |                                     |
|-----------------------------------------|-------------------------------------|
| Short name                              | obs                                 |
| Location                                | 53.99°N, 105.12°W                   |
| Elevation                               | 629 m                               |
| Canopy height                           | 11 m                                |
| Leaf area index                         | 3.5 – 3.8                           |
| Snow-free albedo (above canopy)         | 0.08                                |
| Simulation period                       | 1 October 1997 to 30 September 2010 |
| Temperature/humidity measurement height | 25 m                                |
| Wind measurement height                 | 26 m                                |
| Reference                               | Bartlett et al. (2007)              |

### 5.3 BERMS Old Jack Pine, Saskatchewan, Canada

Jack pine with a sparse understorey, naturally regenerated after fire in 1915. Coarse sandy soil.

**Table 5.3:** Old Jack Pine site characteristics

|                                         |                                     |
|-----------------------------------------|-------------------------------------|
| Short name                              | ojp                                 |
| Location                                | 53.92°N, 104.69°W                   |
| Elevation                               | 579 m                               |
| Canopy height                           | 13 m                                |
| Leaf area index                         | 2.5 – 2.6                           |
| Snow-free albedo (above canopy)         | 0.11                                |
| Simulation period                       | 1 October 1997 to 30 September 2010 |
| Temperature/humidity measurement height | 28 m                                |
| Wind measurement height                 | 29 m                                |
| Reference                               | Bartlett et al. (2007)              |

### 5.4 Col de Porte, France

Grassy meadow bordered by coniferous forest. Soils are 30% clay, 60% sand and 10% silt.

**Table 5.4:** Col de Porte site characteristics

|                                         |                                     |
|-----------------------------------------|-------------------------------------|
| Short name                              | cdp                                 |
| Location                                | 45.30°N, 5.77°E                     |
| Elevation                               | 1325 m                              |
| Snow-free albedo                        | 0.2                                 |
| Simulation period                       | 1 October 1994 to 30 September 2014 |
| Temperature/humidity measurement height | 1.5 m above the snow surface        |
| Wind measurement height                 | 10 m                                |
| Reference                               | Morin et al. (2012)                 |

### 5.5 Reynolds Mountain East, Idaho, USA

Sheltered grassy site in an aspen and fir grove. High organic content in surface soils and 50 cm of silty clay (42% clay, 42% silt) overlying clay (50% clay, 40% silt).

**Table 5.5:** Reynolds Mountain East site characteristics

|                                         |                                     |
|-----------------------------------------|-------------------------------------|
| Short name                              | rme                                 |
| Location                                | 43.19°N, 116.78°W                   |
| Elevation                               | 2060 m                              |
| Simulation period                       | 1 October 1988 to 30 September 2008 |
| Temperature/humidity measurement height | 3 m                                 |
| Wind measurement height                 | 3 m                                 |
| Reference                               | Reba et al. (2011)                  |

## 5.6 Sapporo, Japan

Urban short grass site with frequent deposition of atmospheric aerosols. Clay soil. In situ measurements have been complemented by reanalyses between May and October to form a continuous forcing dataset.

**Table 5.6:** Sapporo site characteristics

|                                         |                                     |
|-----------------------------------------|-------------------------------------|
| Short name                              | sap                                 |
| Location                                | 43.08°N, 141.34°E                   |
| Elevation                               | 15 m                                |
| Simulation period                       | 1 October 2005 to 30 September 2015 |
| Temperature/humidity measurement height | 1.5 m                               |
| Wind measurement height                 | 1.5 m                               |
| Reference                               | Niwano et al. (2012)                |

## 5.7 Senator Beck, Colorado, US

Alpine tundra with thin soils and exposed bedrock.

**Table 5.7:** Senator Beck site characteristics

|                                         |                                     |
|-----------------------------------------|-------------------------------------|
| Short name                              | snb                                 |
| Location                                | 37.91°N, 107.73°W                   |
| Elevation                               | 3714 m                              |
| Snow-free albedo                        | 0.2                                 |
| Simulation period                       | 1 October 2005 to 30 September 2015 |
| Temperature/humidity measurement height | 3.8 m                               |
| Wind measurement height                 | 4.0 m                               |
| Reference                               | Landry et al. (2014)                |

## 5.8 Sodankylä, Finland

Small clearing in a pine plantation with sandy loam soil. Wind speed and radiation are measured above the canopy height, so forcing data have been adjusted for sheltering of the clearing.

**Table 5.8:** Sodankylä site characteristics

|                                         |                                     |
|-----------------------------------------|-------------------------------------|
| Short name                              | sod                                 |
| Location                                | 67.37°N, 26.63°E                    |
| Elevation                               | 179 m                               |
| Simulation period                       | 1 October 2007 to 30 September 2014 |
| Temperature/humidity measurement height | 2 m                                 |
| Wind measurement height                 | 2 m                                 |
| Reference                               | Essery et al. (2016)                |

## 5.9 Swamp Angel, Colorado, USA

Sheltered meadow surrounded by subalpine forest. Colluvium.

**Table 5.9:** Swamp Angel site characteristics

|                                         |                                     |
|-----------------------------------------|-------------------------------------|
| Short name                              | swa                                 |
| Location                                | 37.91°N, 107.71°W                   |
| Elevation                               | 3371 m                              |
| Snow-free albedo                        | 0.2                                 |
| Simulation period                       | 1 October 2005 to 30 September 2015 |
| Temperature/humidity measurement height | 3.4 m                               |
| Wind measurement height                 | 3.8 m                               |
| Reference                               | Landry et al. (2014)                |

## 5.10 Weissfluhjoch, Switzerland

Artificially drained moraine containing clay sediments overlain by serpentine rocks of varying sizes up to 20 cm diameter.

**Table 5.10:** Weissfluhjoch site characteristics

|                                         |                                    |
|-----------------------------------------|------------------------------------|
| Short name                              | wfj                                |
| Location                                | 46.83°N, 9.81°E                    |
| Elevation                               | 2540 m                             |
| Snow-free albedo                        | 0.1                                |
| Simulation period                       | 1 September 1996 to 31 August 2016 |
| Temperature/humidity measurement height | 4.5 m                              |
| Wind measurement height                 | 5.5 m                              |
| Reference                               | WSL (2017)                         |

## 6. Evaluation

Table 6 records the availability of measurements of important snow and soil properties for model evaluation at the reference sites. Basic model evaluation metrics and graphs will be generated automatically and shared with participants. Model outputs will be archived for more sophisticated evaluations. All participants providing site or model data will be included as co-authors on papers using ESM-SnowMIP reference site simulations.

**Table 6:** Evaluation data at reference sites

|                     | cdp | oas | obs | ojp | rme | sap | sbb | sod | swa | wfj |
|---------------------|-----|-----|-----|-----|-----|-----|-----|-----|-----|-----|
| SWE                 | x   | x   | x   | x   | x   | x   | x   | x   | x   | x   |
| Snow depth          | x   | x   | x   | x   | x   | x   | x   | x   | x   | x   |
| Albedo              | x   | x   | x   | x   |     | x   | x   |     | x   | x   |
| Surface temperature | x   | x   | x   | x   |     | x   | x   |     | x   | x   |
| Soil temperature    | x   | x   | x   | x   | x   |     | x   | x   | x   |     |
| Soil moisture       | x   | x   | x   | x   | x   |     | x   | x   | x   |     |

## 7. Submission of results

An ftp address and password for submission of results are available on request from [snowmip@ed.ac.uk](mailto:snowmip@ed.ac.uk). Please submit simulations for reference sites by the end of July 2017.

## 8. References

- Bartlett, P. A., M. D. MacKay and D. Verseghy (2007). Modified snow algorithms in the Canadian Land Surface Scheme: model runs and sensitivity analysis at three boreal forest stands. *Atmosphere-Ocean*, **43**, 207–222.
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- Niwano, M., T. Aoki, K. Kuchiki, M. Hosaka and Y. Kodama (2012). Snow Metamorphism and Albedo Process (SMAP) model for climate studies: Model validation using meteorological and snow impurity data measured at Sapporo, Japan. *J. Geophys. Res.*, **117**, F03008, doi:10.1029/2011JF002239.
- Reba, M. L., D. Marks, M. Seyfried, A. Winstral, M. Kumar and G. Flerchinger (2011). A long-term data set for hydrologic modeling in a snow-dominated mountain catchment. *Water Resour. Res.*, **47**, W07702, doi:10.1029/2010WR010030.
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## Document history

|                 |                                  |
|-----------------|----------------------------------|
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